

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
3 FT	(a)	(i)		Volume of soap used (to form a lather) Do not accept 'amount'	1			1		1
		(ii)		Same water sample to be checked by different technicians/ different technicians will check water samples		1		1		1
	(b)	(i)		Indicative content Hard water areas are Southampton and Bristol. Soft water found in Birmingham and Manchester. Southampton has temporary hard water - so less volume of soap required to form a lather after boiling. Boiling softens temporary hard water. Calcium and/or magnesium hydrogen carbonates break down by boiling. Carbonates form lime-scale 5– 6 marks Identification of cities with hard and soft water and candidates recognise the changes in the hardness of water by the volume of soap required. Candidates recognise that the water from Southampton has become softer as a result of boiling, (temporary hard water) so less soap required to form a lather. The candidates explain the chemical changes occurring in the temporary hard water with the formation of carbonate and lime-scale. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3 – 4 marks Identification of cities with hard and soft water and candidates recognise the changes in the hardness of water by the volume of soap required. Candidates may recognise that the Southampton water has become softer as a result of boiling, (temporary hard water) less soap required to form a lather. They may mention formation of lime-scale.		2	4	6		6

			<i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 2 marks Candidates recognise cities with hard and soft water. OR Candidates recognise the changes in the hardness of water by the volume of soap required. May include a reference to temporary hard water in Southampton. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate used limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks No attempt made or no response worthy of credit.						
		(ii)	Any 2 × (1) from: furring or lime-scale in a kettle (1) (lime-scale causing) blocked pipes (1) inefficient boilers(1) scum with soaps/uses a lot of soap/can't get bubbles/can't get lather (1) decreased lifetime of appliances (1)	2					
			Question 3 total	3	3	4	10	0	8